

Final Assignment: Triton ResNet

MGP 2025 Spring

Due: 08 JUN (SUN) 23:59

No Late Submission For This Assignment

Writing custom GPU kernels is not free. While we often rely on PyTorch and implicitly leverage high-performance kernels from CUDA libraries (such as cuBLAS, cuDNN, and CUTLASS), there are scenarios where crafting specialized kernels is advantageous. Although library kernels are typically of very high quality and straightforward to use, this assignment challenges you to delve deeper by implementing ResNet18 manually in Triton.

[Overview]

You are required to implement the Triton kernels in the `TritonMGP/kernel` directory. During grading, only the `TritonMGP/kernel` directory will be copied to the grading server.

First, run `make info` to get details about the ResNet18. Next, run `make torch` to see the reference PyTorch results. Then run `make triton`, and this is the target you need to complete by filling in the Triton kernels. The file `TritonReLU_ResNet18.py` serves as a hint: it shows how `mgp_nn.TritonReLU` is implemented. Finally, run `make test-triton-relu` to test your ReLU kernel and observe its output. At the moment, this model only replaces the ReLU activation with a Triton implementation. Please note that until you've implemented all of the kernels in `~/mgp/hw6`, running `make test-triton-relu` only works in the `~/mgp/hw6-example` directory.

Your ResNet18 model must achieve **at least 75% accuracy**.

You cannot use any torch functions in this assignment; you may only use the `empty`, `empty_like`, `zeros`, `ones`, `randn` and `rand` functions from the `mgp` package. See the implementation of `mgp_nn.TritonReLU` for an example of how to use them.

[Code - 10 pts]

The total is 10 pts.

- 10 pts, if your model works within 500ms. No partial points.

Bonus points are awarded if your program outperforms.

- 5pt, if your model works faster than 200ms.

[Report - 10 pts]

You have to submit an "excellent" report about your homework including why Conv2d does not show the efficient performance compared to torch. Your report grade will **vary significantly** depending on its quality. Any use of GPT for this particular report will incur a **substantial penalty**.

[Note]

Your code's speed is measured based on the maximum of 5 executions on the grading server. I won't regrade due to "It worked on my machine" excuses, so I encourage you to perform the assignment more generously than the criteria. As many students tend to rush near the deadline, your code may appear slower in the local environment. Therefore, I recommend completing the assignment **as early as possible** when the server is less congested.

[Rules]

1. Put your HW6 directory under <your_home>/mgp/hw6
 - a. All your make commands, especially 'make run' should work.
2. **Do not change any** of the output(i.e. 'std::cout') statements. We will grade based on the program outputs. Any change on the outputs would probably cause an error, leading to 0 points for your HW assignment.
3. Don't touch anything in the directory after the deadline. If any of the timestamps are later than the deadline, we will treat it as 'late submission'.
4. You can't use CUDA libraries for this assignment.
5. Please retain adequate permission on your files to avoid plagiarism.
6. No abusing. Please use your common sense.
7. Zero GPT policy. No mercy